EXECUTIVE ENGINEERING WORK REPORT

LeaderApp Mobile Engineering — Complete API Integration, Features & Bug Remediation

Reporting Period:	Last 13 Days (Oct 2 – Oct 15, 2026)	**Application:**	LeaderApp (Executive Suite)
Architecture:	MVP + BLoC (State Machine)	**Static Analysis:**	**0 Errors, 0 Warnings**
Primary Tech:	Flutter 3.x / Dart / REST APIs	**Backend Scope:**	Dev / Staging / Production Environments

1. Executive Summary

Over the past 13 days, the LeaderApp mobile engineering team completed an extensive architectural overhaul, full live API integration, and user authentication stabilization. All mock/dummy fixtures across critical modules (Reports, Peers, Teams, and Analytics) have been eradicated and replaced with real-time API integrations, strict MVP (Model-View-Presenter) view decoupling, and resilient BLoC state machines. The app now compiles flawlessly with zero warnings and zero errors across the entire codebase.

2. Comprehensive API Integration & Modification Register

The table below details all 25 REST API endpoints integrated, aligned, or updated during this cycle:

#	Module	Method & Endpoint	Status	Description / Payloads / Fixes
1	Auth	`POST /api/v1/leader/auth/send-otp`	**Updated/Live**	Sends OTP to leader email or mobile ({'email_or_phone': '...'}).
2	Auth	`POST /api/v1/leader/auth/verify-otp`	**Updated/Live**	Verifies OTP and returns leader JWT auth token & profile.
3	Profile	`GET /api/v1/leader/profile`	**Live**	Retrieves active leader role, capabilities matrix & credentials.
4	Peers	`GET /api/v1/peers`	**Live**	Peer directory listing with query search, circle & status filters.
5	Peers	`GET /api/v1/peers/:id`	**Live**	Detailed peer profile, parsed life_impacted_count, contact details.
6	Peers	`GET /api/v1/peers/celebrations`	**Live**	Live member birthday and work anniversary calendar.
7	Peers	`POST /api/v1/peers/:id/send-wish`	**Live**	Sends greetings/wishes to peer members for celebrations.
8	Teams	`GET /api/v1/teams/circles`	**Updated/Live**	Circle directory. Added support for dynamic circle_image parsing.
9	Teams	`GET /api/v1/teams/industries`	**Live**	Dynamic parent industry categories and circle cluster totals.
10	Teams	`GET /api/v1/teams/summary`	**Live**	Summary metrics across circles (Total Circles, Health, Revenue).
11	Circle	`GET /api/v1/teams/circles/:id`	**Live**	Single circle executive management details, chairs & founders.
12	Circle	`GET /api/v1/teams/circles/:id/peers`	**Live**	Dedicated peer roster belonging strictly to selected circle.
13	Circle	`GET /api/v1/teams/circles/:id/sub-industries`	**Live**	Active and open sub-industry slots per circle.
14	Circle	`GET /api/v1/leader/teams/circles/:id/events`	**Live**	Circle-scoped assemblies & summits (?filter=all|upcoming).
15	Reports	`GET /api/v1/leader/reports`	**Live**	Historical verified report submissions and status audit trail.
16	Reports	`GET /api/v1/leader/reports/attendance-trend`	**Live**	Live attendance analytics, meeting trends & ratios over time.
17	Reports	`GET /api/v1/leader/reports/export`	**Live**	Server-side triggered data exports for leadership records.
18	Member 360	`GET /api/v1/leader/members/:id`	**Live**	Comprehensive 360-degree overview of individual circle member.
19	Member 360	`GET /api/v1/leader/members/:id/activities`	**Live**	Consolidated member activity feed (deals, meetings, referrals).
20	Member 360	`GET /api/v1/leader/members/:id/events`	**Live**	Member registered assemblies, passes, and attendance status.
21	Leader Asks	`GET /api/v1/leader/asks/collaboration/my`	**Live**	Leader collaboration asks pipeline and fulfillment tracking.
22	Leader Asks	`GET /api/v1/leader/asks/referral/my`	**Live**	Business referral pipeline and partner introductions.
23	Leader Asks	`GET /api/v1/leader/asks/help/my`	**Live**	Peer community advisory and urgent help channels.
24	Finance	`GET /api/v1/finance/metrics`	**Live**	Circle collection metrics, transaction yields & pending amounts.
25	Finance	`GET /api/v1/finance/transactions`	**Live**	Live transaction audit trail with verified/pending status tracking.

3. Core Modules Delivered & Enhancements

A. Authentication & Release APK Environment Resolution

• **Dedicated Leader Authentication:** Fully integrated endpoints /leader/auth/send-otp and /leader/auth/verify-otp, ensuring leader role privileges are validated on login.

• **APK Environment Conflict Resolved:** Fixed the *'User not found'* error that occurred on generated release APKs. The app previously toggled to production in release mode; it now safely defaults to dev.peersunity.com with optional --dart-define=ENV=prod override for production builds.

• **Secure Session Persistence:** JWT bearer tokens, active roles, and capability profiles are reliably persisted and refreshed across app restarts via SessionManager.

B. Reports & Analytics (100% Live Data & Full Export Engines)

• **Mock Data Eradication:** Replaced all placeholder fixtures in ReportsView, ReportHistoryCard, and ReportsExportSection with live REST calls.

• **View Full Report & Peer Roster:** Implemented real-time peer roster synchronization via /peers?circle_id=... with live search, industry badges, and attendance analytics.

• **Client-Side PDF Document Generator:** Built a production-quality PDF export engine generating formatted leadership reports, executive metrics summaries, and tabular rosters.

• **Excel (.csv) Export Engine:** Implemented RFC-4180 compliant CSV spreadsheet generation with automatic comma/quote escaping and direct native device sharing via share_plus.

C. Circles & Teams UI/UX Overhaul

• **Dynamic Cover Banner Image:** Added support for backend circle_image payloads, rendering images with BoxFit.cover within top 16px rounded card borders without distorting or clipping critical edges.

• **Avatar Badge Removal:** Removed the overlapping circular initials avatar badge (e.g. 'AD', 'AE', 'AF') that was obstructing the bottom-left corner of cover images.

• **Balanced Visual Hierarchy:** Redesigned card body padding (12px top) for an executive, modern presentation.

D. Universal Case-Insensitive & Multi-Field Search Engine

• **Universal Case-Insensitivity:** Eliminated bugs where typing lowercase letters returned empty search results.

• **Multi-Field Querying:** Circle search now queries Name, Location, Category/Industry, Chairs, Founders, Directors, dynamic Tags, and word acronyms/initials (e.g. 'ad' matches 'Ahmedabad District').

• **Keyboard Optimization:** Configured textCapitalization: TextCapitalization.none and TextInputAction.search to prevent mobile keyboards from enforcing uppercase characters.

E. Peers Directory & Live Impact Score

• **Lives Impacted Binding:** Parsed the backend's life_impacted_count: 222 JSON field, accurately binding lives impacted to peer cards, profile sheets, and coins/leaderboards.

• **Celebrations & Greetings:** Integrated live member birthdays and anniversaries feed with instant one-tap greeting wishes dispatch.

4. Technical Errors Encountered & Engineering Resolutions

Error 1: 'User not found' on Release APK Login

Root Cause: When building release APKs via flutter build apk, Flutter sets kDebugMode = false. In api_endpoints.dart, activeEnvironment previously defaulted to AppEnvironment.prod (production URL) when kDebugMode was false. Development test accounts only exist on dev.peersunity.com, causing the production server to reject login.
Resolution: Updated activeEnvironment to default to dev.peersunity.com across all debug and release builds. Added compile-time --dart-define=ENV=prod flag for explicit production deployments.

Error 2: Search Failing with Lowercase Input

Root Cause: In TeamsBloc, search was restricted to basic substring queries without acronym or tag matching. Concurrently, peer profile ask screens checked !title.contains(_searchQuery) without lowercasing the source title, causing all lowercase queries to fail against capitalized titles.
Resolution: Upgraded _filterAndSortCircles with tokenized, acronym-aware, case-insensitive logic. Normalized all comparative fields with .toLowerCase().trim().

Error 3: Missing Impact Count on Peer Cards

Root Cause: UI displayed 0 or fallback values because the model previously anticipated impact_score, while the backend delivers life_impacted_count.
Resolution: Updated PeerModel.fromJson to prioritize life_impacted_count with safe numeric parsing and legacy fallback.

Error 4: Non-Functional Export Buttons (PDF & Excel)

Root Cause: Export actions contained placeholder stubs and lacked file system writing permissions and mobile sharing intents.
Resolution: Implemented native document generators for CSV and PDF, integrated share_plus to invoke native iOS/Android sharing sheets, and formatted data with timestamps and audit headers.

Error 5: Memory Leak & Controller Rebuild Loss

Root Cause: TextEditingController was instantiated directly inside a StatelessWidget (_TeamsContent). Parent state rebuilds created brand new controller instances, clearing user input and leaking memory in violation of Rulebook Rule 4.
Resolution: Converted _TeamsContent to a StatefulWidget, initializing in initState() and disposing in dispose().

5. Architectural Adherence & Code Quality

• **Purity of Pattern:** 100% adherence to MVP (Model-View-Presenter) with BLoC (flutter_bloc). Feature isolation maintained in lib/features/[feature_name]/.
• **Compiler Health:** Executed flutter analyze project-wide — **0 errors, 0 warnings**.
• **Role-Based Access Control (RBAC):** Dynamic layout customization based on active role (Circle Chair, Founder, Director, Super Admin) with fallback views for restricted tiers.
• **Resource Hygiene:** All scroll controllers, search controllers, and stream subscriptions are strictly disposed in widget lifecycle hooks.

6. Next Steps & Recommended Milestones

1. **Physical Device QA Sign-off:** Complete regression testing on the generated development APK on Android and iOS devices.
2. **Push Notification Integration:** Verify background FCM message receipt for circle assemblies and peer celebration alerts.
3. **Production Staging:** Finalize production release pipeline using flutter build apk --dart-define=ENV=prod.

Prepared By: Lead Mobile Engineer	**Approved By:** Engineering Director	**Date:** October 15, 2026